Make Up Class

Monday, 13 April 2026 5:06 pm

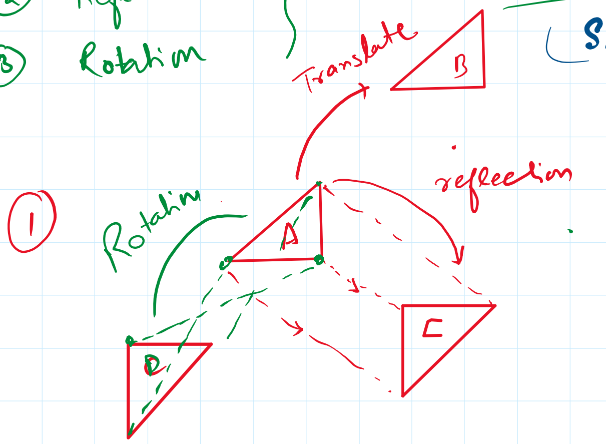
· Transformation to Shift Co-ordinates (object) under certain condition

$$\underset{\text{matrix}}{A} X = \underset{\text{image}}{X'}$$

New Co-ordinate (Image

1. Translation
2. Reflection
3. Rotation

} Euclidean Transformation (Shape & Size is not change)



Translation column matrix $\begin{pmatrix} a \\ b \end{pmatrix}$

Rotation & Reflection $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$AX = X'$$

$$X = A^{-1} X'$$

4.9 Matrix Transformations from R^n to R^m

A *matrix transformation* T: $R^n \to R^m$ is a mapping of the form

$$\mathbf{T}(\vec{x}) = \mathbf{A}\vec{x},$$

$A\vec{x} = x'$

for all $\vec{x}$ vectors in R^n and an $m \times n$ matrix A.

R^n R^m

The matrix transformations are precisely the *linear transformations* from R^n to R^m, that is, the transformations with the linearity properties

$T(u + v) = T(u) + T(v)$ and $T(ku) = kT(u)$

We will use these two properties as the starting point for defining more general linear transformations.

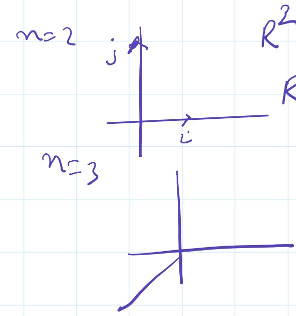
$n=1$

$n=2$

$R^2 \to R^3$

$R^2 \to R^1$

$R^3 — R^2$



Remark: It is important to note that a linear transformation is a special kind of function. The input and output are both vectors.

If we denote the output vector $\mathbf{T}(\vec{x})$ by $\vec{y}$ we can write

$$\vec{y} = \mathbf{A}\vec{x}$$

Finding the Standard Matrix for a Matrix Transformation

Step 1. Find the images of the standard basis vectors $e_1, e_2, \ldots, e_n$ for R^n in column form.

Step 2. Construct the matrix that has the images obtained in Step 1 as its successive columns. This matrix is the standard matrix for the transformation.

$n=3$

Types of linear Transformations

There are two types of linear transformations (defining from R^2 to R^2):

1- Euclidean Transformation
2- Affine Transformation

$T\begin{pmatrix} 3 \\ 3 \end{pmatrix} + T\begin{pmatrix} 3 \\ 2 \end{pmatrix} = T\begin{pmatrix} 5 \\ 5 \end{pmatrix}$

1 – Euclidean Transformations

A Euclidean Transformation is a transformation T: $R^2 \to R^2$ defined by

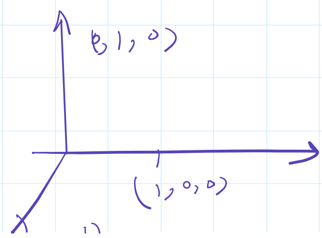
$$\mathbf{T}(x) = \mathbf{A}\vec{x} + \vec{a}, \quad \forall \vec{x} \in R^2$$

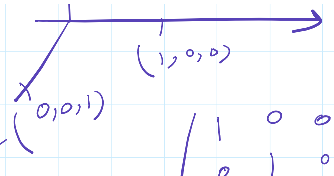
$T(5u) = 5\,T(u)$

Where A is an orthogonal 2 x 2 matrix and $\vec{a} \in R^2$. These types of transformations always preserve distance/shape.

An orthogonal matrix holds the property $AA^T = I$ or $A^T = A^{-1}$

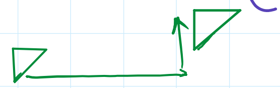
$$A^T = A^{-1}$$





$$z \; (0,0,1) \qquad (1,0,0)$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



$$T + O = I$$

$$\boxed{TPOEI}$$

1- Translation

Translation is a transformation from (R^2 to R^2) or (R^3 to R^3) defined as:

$$T(\vec{x}) = \vec{x} + \vec{a}, \qquad \forall \vec{x} \in R^2$$

where matrix A is the identity matrix.

Example 1: (Translation of a triangle)

Let A = (-2, -2), B = (2, -2), C = (0, 2) form a triangle. Find the translated triangle with vector $\vec{a} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$.

$$D = A + \vec{a} = \begin{pmatrix} -2 \\ -2 \end{pmatrix} + \begin{pmatrix} 5 \\ 8 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

$$E = B + a = \begin{pmatrix} 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 5 \\ 8 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$$

$$F = C + a = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 5 \\ 8 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$$

Called as Translation of Co-ordinates

$(2, -3)$, $I = (4, 6)$

$(2, -3)$ is Translated to $(4, 6)$ find Translation vector.

$$\begin{pmatrix} 2 \\ -3 \end{pmatrix} + T = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

$$T = \begin{pmatrix} 4 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 4-2 \\ 6+3 \end{pmatrix} = \begin{pmatrix} 2 \\ 9 \end{pmatrix}$$

Translation of function

Linear $\boxed{2x + 3y = 10}$ object

Translate it in $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

$$x' = x - 2 \quad \Rightarrow \quad x = x' + 2$$

$$y' = y + 3 \quad \Rightarrow \quad y = y' - 3$$

$$2(x' + 2) + 3(y' - 3) = 10$$

$$2x' + 4 + 3y' - 9 = 10$$

$$2x' + 3y' - 5 = 10$$

Im $\boxed{2x' + 3y' = 15}$

① let $(x-2)^2 + (y+3)^2 = 49$

$$(x - x_1)^2 + (y - y_1)^2 = r^2$$

$x_1 = 2$

$y_1 = -3$

$r = 7$

$$(x-x_1)^2 + (y-y_1)^2 = r^2 \qquad y_1 = -3,\ r = 7$$

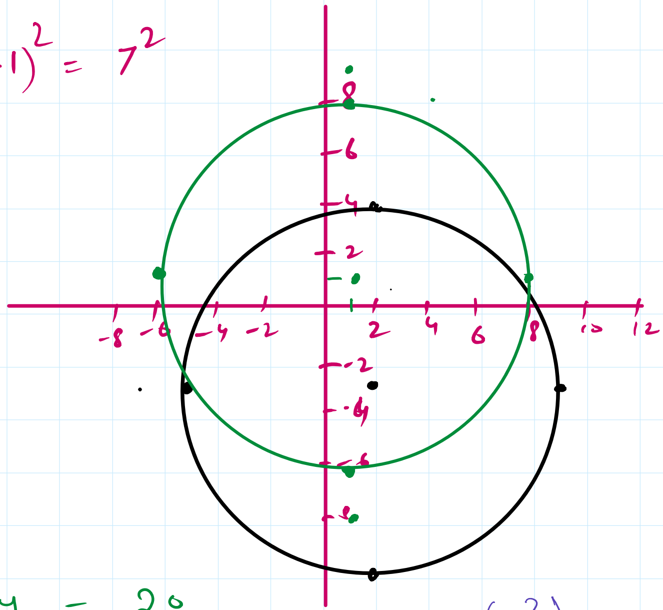
Translate in $\begin{pmatrix} -1 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$x' = x-1 \qquad \Rightarrow x = x'+1$

$y' = y+4 \qquad y = y'-4$

$$(x'+1-2)^2 + (y'-4+3)^2 = 7^2$$

$$(x'-1)^2 + (y'-1)^2 = 7^2$$



$$x^2 - 4x + y^2 + 6y = 20$$

$\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$$x^2 - 4x + 2^2 + y^2 + 6y + 3^2 = 20 + 2^2 + 3^2$$

$$(x-2)^2 + (y+3)^2 = 20 + 4 + 9 = 33$$

$x' = x+2 \qquad \Rightarrow x = x'-2$

$y' = y+3 \qquad y = y'-3$

$$(x-2)^2 + (y+3)^2 = (\sqrt{33})^2$$

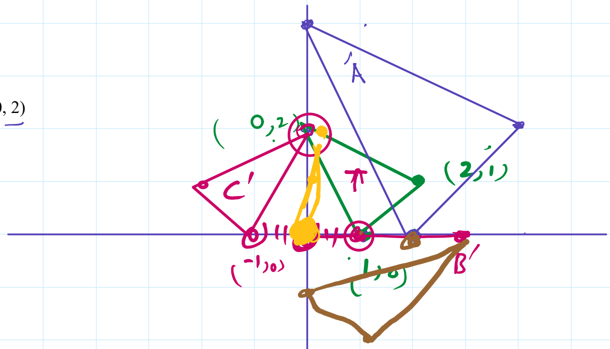
Work to do:

Q1. Consider the letter L in figure, made up of the vectors (1, 0) or $\begin{bmatrix}1\\0\end{bmatrix}$ and (0, 2) or $\begin{bmatrix}0\\2\end{bmatrix}$, show that the effect of the linear transformation on the matrices and describe each of the transformation in words.

A $= \begin{bmatrix}2 & 0\\0 & 2\end{bmatrix}$, B $= \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$,

C $= \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}$, D $= \begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$,

E $= \begin{bmatrix}0 & 0.2\\0 & 1\end{bmatrix}$, F $= \begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}$



① $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 4 \\ 0 & 4 & 2 \end{pmatrix}$

$R^2 \to R'$ by

① $\begin{pmatrix} \cdot & \cdot \\ 0.2 & 2 \end{pmatrix}\begin{pmatrix} 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 4 & 2 \end{pmatrix}$ by

② B) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$

③ C) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & -2 \\ 0 & 2 & 1 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 0 & -2 \end{pmatrix}$$

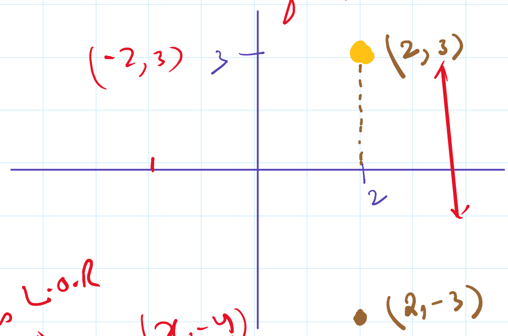
$$\begin{pmatrix} 0 & 0.2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 0 & -2 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0.4 & 0.2 \\ 0 & 2 & 1 \end{pmatrix}$$

Reflection Line of Reflection

① x-axis (change sign of y-coord) ③ $y=x$

② y-axis (change sign of x-coord) ④ $y=-x$

$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$



$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

(x, y)

x-axis L.O.R or $y=0$ → $(x, -y)$

y-axis L.O.R $x=0$ → $(-x, y)$

$y=x$ → (y, x) (interchange)

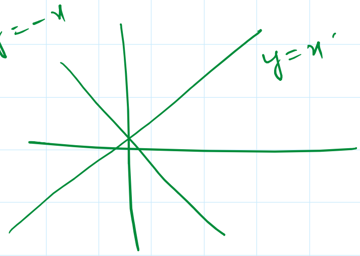
$y=-x$ → $(-y, -x)$

$y=x$

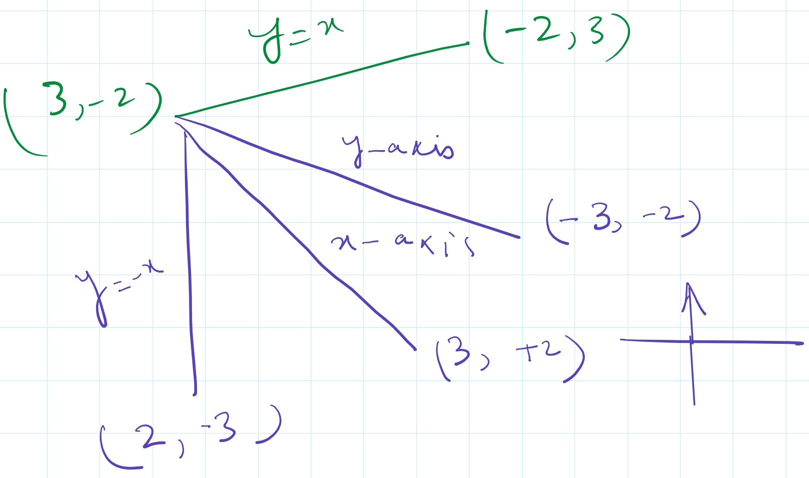
$(1, 4)$ $(4, 1)$

$(0, 8) \Rightarrow$ $y=x$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$



$y=-x$ $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$



L.O.R	Matrix
X-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
$y=x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
$y=-x$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$$T(x) = AX$$

Co-ordinate $\begin{pmatrix} 4 & 2 & -3 \\ -2 & 3 & 2 \end{pmatrix}$

Reflected in line $y=0$ (x-axis) $= \begin{pmatrix} 4 & 2 & -3 \\ 2 & -3 & 2 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 2 & -3 \\ -2 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 2 & -3 \\ 2 & -3 & -2 \end{pmatrix}$$

$y=x$ $\begin{pmatrix} -2 & 3 & 2 \\ 4 & 2 & -3 \end{pmatrix}$

① $2x + 3y = 5$, Reflect it in $y=x$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$y = x'$

$x = y'$

$$2y' + 3x' = 5$$

$$3x' + 2y' = 5$$

$$3x' + 2y' = 5$$

$$(x-2)^2 + (y+5)^2 = 5^2$$

reflect in line y-axis $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix} \Rightarrow \begin{array}{l} x = -x' \\ y = y' \end{array}$$

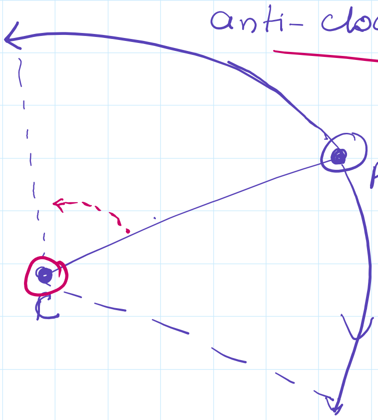
$$(-x'-2)^2 + (y'+5)^2 = 5^2$$

$$(x+2)^2 + (y'+5)^2 = 5^2$$

Rotation

- Centre of Rotation $(0,0)$
- angle of Rotation
- direction

anti-clock wise $\theta = +ve$

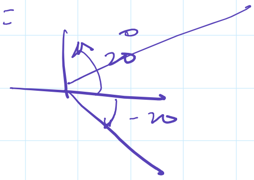


Clock wise $\theta = -ve$

$\sin(\theta) =$

$\sin 20$

$\sin -20$



$\sin(-\theta)$

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$AX = X'$$

$\theta =$ Anti-clock

$$\boxed{A = \begin{bmatrix} Cos(\theta) & -Sin(\theta) \\ Sin(\theta) & Cos(\theta) \end{bmatrix}}$$

$\theta =$ clockwise

$\theta = -\theta$

$$\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

Clock wise Rotation

$$T(e_1) = T(1,0) = (\cos\theta, -\sin\theta)$$

$$T(e_2) = T(0,1) = (\sin\theta, \cos\theta)$$

$T(e_2) = T(0,1) = (\sin\theta, \cos\theta)$

① Rotat $(2,3)$ $\theta = 30^\circ$ anticlock

$$\begin{pmatrix} \cos 30 & -\sin 30 \\ \sin 30 & \cos 30 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} =$$

$\theta = 90$
$\theta = 180$
$\theta = 270$

$$\begin{pmatrix} \cos 90 & -\sin 90 \\ \sin 90 & \cos 90 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} =$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

Q Rotat $2x + 3y = 10$ Thru 270 Anticlw

$$\begin{pmatrix} \cos 270 & -\sin 270 \\ \sin 270 & \cos 270 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\begin{pmatrix} y \\ -x \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$y = x'$
$-x = y'$
$x = -y'$

$2(-y') + 3x' = 10$
$-2y' + 3x' = 10$
$3x' - 2y' = 10$

Inversion Algorithm to find Inverse of a matrix